



STEP 5 — APPOINTMENT BOOKING SYSTEM (GHL Calendar Clone)

This feature is **VERY** powerful for real clients like:

- Doctors
 - Coaches
 - Agencies
 - Wedding services (perfect for your niche 🔥)
-



STEP 5 GOAL

Create a system where:

- Clients can book appointments from a link
 - Staff availability is managed
 - Time slots are auto-generated
 - Confirmations + reminders are automated
-



5.1 ARCHITECTURE OVERVIEW

Booking Page (Public)



Select Date & Time



Check Availability



Create Appointment



Save in CRM



Trigger Automation (WhatsApp / Email)



5.2 DATABASE DESIGN

✓ Table: **tbl_calendars**

```
CREATE TABLE tbl_calendars (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(191),  
  staff_id INT,  
  slot_duration INT DEFAULT 30, -- minutes  
  buffer_time INT DEFAULT 0,  
  max_bookings_per_day INT DEFAULT NULL,  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

✓ Table: **tbl_calendar_availability**

```
CREATE TABLE tbl_calendar_availability (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  calendar_id INT,  
  day_of_week INT, -- 0=Sunday, 6=Saturday  
  start_time TIME,  
  end_time TIME  
);
```

✓ Table: **tbl_appointments**

```
CREATE TABLE tbl_appointments (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  calendar_id INT,  
  lead_id INT,  
  staff_id INT,  
  appointment_datetime DATETIME,  
  status VARCHAR(50) DEFAULT 'booked',  
  notes TEXT,  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```



5.3 BOOKING PAGE (PUBLIC)

✓ URL Example

https://yourcrm.com/book/{calendar_slug}

UI Layout

Select Date



Available Time Slots



Enter Details (Name, Phone, Email)



Confirm Booking



5.4 TIME SLOT GENERATION

Logic

Example:

- Working hours: 10:00 → 18:00
 - Slot duration: 30 mins
-

PHP Function

```
function generate_slots($start, $end, $duration)
{
    $slots = [];

    $start_time = strtotime($start);
    $end_time = strtotime($end);

    while ($start_time < $end_time) {
        $slots[] = date('H:i', $start_time);
        $start_time += ($duration * 60);
    }

    return $slots;
}
```

5.5 REMOVE BOOKED SLOTS

✓ Query

```
$this->db->where('DATE(appointment_datetime)', $selected_date);  
$this->db->where('calendar_id', $calendar_id);  
$bookings = $this->db->get('tbl_appointments')->result();
```

Logic:

- Remove already booked slots
 - Apply buffer time if needed
-

5.6 BOOK APPOINTMENT (API)

✓ Controller

```
public function book_appointment()  
{  
    $data = $this->input->post();  
  
    // Create or find lead  
    $lead = $this->db->where('phonenumber', $data['phone'])->get('tblleads')->row();  
  
    if (!$lead) {  
        $this->db->insert('tblleads', [  
            'name' => $data['name'],  
            'email' => $data['email'],  
            'phonenumber' => $data['phone']  
        ]);  
        $lead_id = $this->db->insert_id();  
    } else {  
        $lead_id = $lead->id;  
    }  
  
    // Save appointment  
    $this->db->insert('tbl_appointments', [  
        'calendar_id' => $data['calendar_id'],  
        'lead_id' => $lead_id,  
        'staff_id' => $data['staff_id'],  
        'appointment_datetime' => $data['datetime']  
    ]);
```

```
// Trigger automation
trigger_event('appointment_booked', [
  'lead_id' => $lead_id,
  'datetime' => $data['datetime']
]);

echo json_encode(['status' => 'booked']);
}
```



5.7 CONFIRMATION + REMINDERS



After Booking:

Trigger:

```
trigger_event('appointment_booked', ...)
```

n8n Workflow:

1. Send WhatsApp:
 - “Your appointment is confirmed”
 2. Wait until:
 - 1 hour before appointment
 3. Send Reminder:
 - “Reminder: Your appointment at 3 PM”
-



5.8 RESCHEDULE / CANCEL



Add Fields:

```
ALTER TABLE tbl_appointments
ADD rescheduled_from DATETIME NULL,
ADD cancelled_at DATETIME NULL;
```



API:

```
public function cancel_appointment($id)
{
    $this->db->where('id', $id);
    $this->db->update('tbl_appointments', [
        'status' => 'cancelled',
        'cancelled_at' => date('Y-m-d H:i:s')
    ]);
}
```



5.9 ADMIN CALENDAR VIEW



Page:

/admin/calendar

Options:

◆ **Simple:**

- List view

◆ **Advanced:**

Use:

- FullCalendar JS
-

Features:

- Daily view
 - Weekly view
 - Drag to reschedule
-



5.10 GOOGLE CALENDAR SYNC (ADVANCED)

Flow:

- Booking → Push to Google Calendar
 - Google → Sync back (optional)
-

Use:

- Google Calendar API
-

5.11 DOUBLE BOOKING PROTECTION

Check Before Insert:

```
$this->db->where('appointment_datetime', $data['datetime']);  
$this->db->where('staff_id', $data['staff_id']);  
$exists = $this->db->get('tbl_appointments')->row();  
  
if ($exists) {  
    echo json_encode(['status' => 'error', 'message' => 'Slot already booked']);  
    exit;  
}
```

5.12 TESTING CHECKLIST

Must Work:

- Time slots generate correctly
- Book appointment
- No double booking
- Lead auto-created
- Automation triggered

- Reminder working
 - Cancel/reschedule works
 - Admin calendar shows data
-



STEP 5 OUTPUT

After this:

- ✓ Booking system (like Calendly/GHL)
- ✓ Lead + appointment connected
- ✓ Reminder automation
- ✓ Calendar UI
- ✓ Real business use-case ready